

TCP/IP Model-IP Addressing - II

Module 10

Duration: 28:44

Video: <https://www.youtube.com/watch?v=FQzPOhnEkjg>

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0:00 hello class in the previous lecture we had talked about the flow of data from a
0:08 application layer to the network layer now we said that from application layer the
payload gets transported to
0:17 transport layer where the segment is formed and then it is transported to the
network layer where a packet is
0:26 formed now this packet is formed with the help of two two things one is called The
Source IP address and other is called the destination IP
0:34 address so we discussed about the various classes of IP addresses and we talked
about class A IP
0:42 addresses Class B and Class C and we have taken one example where we wanted
0:48 to have a network establishment of a 20 machines so we are going to choose an IP
0:55 [Music]
0:58 192.168.1.0 and I want to address all the machines in my network where the
wastage would be as
1:05 less as possible so we are going to utilize cidr which is called classless interdomain
1:13 routing and this classless Internet Internet domain routing is going to prove
sufficient enough in order to
1:22 lessen the wastage of IP addresses so
1:28 192.168.1.0 is our Network address and in order to address 20 machines we'll
1:37 require five bits now how we come to this five bits so you need to memorize a
1:44 table which is called 2^0 to 2^7 because computer is
1:52 going to use a binary system 2^0 to 2^7 now if you convert into the
2:00 decimal format this is 1 2 4 8 16 32 64
2:10 128 and it will go 2^7 to 8 and so and so forth now these are the eight bits of
2:19 the last octet so out of these one 2 3 4
2:25 5 these five bits we want that that these should address host and these
2:32 three bits should address the network now wherever the weight is one
2:39 that will represent Network and wherever the weight is zero that will represent

2:46 host so now we'll add up these values 128 64 and
3:54 32 so this comes out to be 224 that means the last octet in
3:01 3 minutes, 1 second our case is 255 do 255 do 255 do
3:10 224 now this 224 is going to give us five bits as
3:19 hosts so our address is 192.168.1.0 is our Network address the
3:27 first machine will be 192 168 1 1 second will be 1.2
3:35 1.3 and it will go up to 1.20 so these are the 20 machines which will be on our
3:43 Network now the thing comes what would be the broadcast address of my network
now how to calculate the
3:50 broadcast address so we are going to write 192 168 do
3:56 [Music]
3:58 1.0 and subnet mask is 255 255.255
4:06 224 now whenever we have to do some calculations on IP addresses always
remember that first of all take the
4:14 incomplete block so here the incomplete block is 224 so you subtract this from
4:22 256 so this comes out to be 32 so the block size
4:30 is 32 that means in each block there will be 32 machines which can get address
from this IP
4:39 scheme now the last address from this will be the broadcast so if you say 32 -
4:46 1 this comes out to be 31 so that means 192
4:53 168 do 1.31 is the broadcast address
5:02 192.168.1.0 is the network address from 21 to address 30 this is
5:10 the wastage because we only required 20 machines so this is how we are going to
5:18 calculate the number of machines on a network what is the network address what
is the broadcast let's take another example
5:26 [Music]
5:28 192.168.1 0 this is the network address
5:36 calculate broadcast address first host
5:46 address last host address number of
5:55 hosts and number of
6:05 subnets if we want to have 10 machines on the
6:14 network now let's do this so we have been given address 192 168
6:22 1.0 now we have to address how many machines 10 machines so let's again draw
that table
6:29 1 2 4 8 16 32 64 128 so total there are
6:37 1 2 3 4 5 6 7 8 bits so these are the eight
6:44 bits which will address 256 machines so we need to address only 10 machines so
6:52 in order to address 10 machines I would require four bits so four bits are required

for addressing this so that

7:00 means these four bits will represent Network and these four bits will represent
7:09 host so 2 to 4 - 2 is the number of
7:17 hosts which comes out to be 14 so we'll be able to address 14 machines with help
of this addressing
7:25 mechanism so what would be my subnet mask let's calculate so my subnet mask is
7:32 255 do 255 do 255 dot 1 2 3 4 4 1es and four
7:42 zeros so if we convert this into decimal this comes out to be 240 so all the weights
which are being
7:51 given one value you need to add
7:54 those so 1 2 4 8 16 32 64 128 1 1 1 1 0
8:05 0 0 0 so these values are to be added so $128 + 64 + 32 + 16$ this comes out to be
8:16 240 so this 240 is my last octet so your last OCT dat is
8:23 240 so the address will be 192.168.1.0 and the subnet mass will be
8:31 255 do 255.255 do 2 40 so that means I am going
8:40 to consume four bits for host and 28 bits for
8:48 Network because my IP V4 address is of 32 bits so out of these 32 bits only
8:57 four bits are being used as host and rest 28 are being used for Network so we can
also specify in
9:06 another format 192.168.1.0 /28 so this is called
9:15 cidr notation so IP address can either be specified in a subnet MK mask notation
9:24 or they can be specified as cidr notation one and same thing so our network is
9:35 192.168.1.0 /28 so this 28 is basically 255 do
9:44 255 255 240 now let's calculate the block so
9:51 block is calculated with the help of incomplete octet so here the fourth OCT is
incomplete
10:00 rest all are complete 255 is the last value which this octet can have so
10:07 subtract it from 256 so 256 minus 240 this gives me the value 16 so 16 is my
10:16 block size so the first address in the block is my network address
10:26 192.168.1.0 this is the first address and the last address in the block will be my
broadcast
10:34 address which is 192.168.1.5 so these are the addresses
10:45 for my network so 192.168.1.0 is the network
10:52 address 192.168.1.5 is the broadcast address
11:02 first machine will have 192.168.1.1 as the address and the last machine will be
11:12 192.168.1.10 because I had only 10 machines in my network but we have few more
addresses
11:21 11 12 13 and 14 so these four addresses are going to get faced if we are going

11:28 to use C IDR notation in order to address 10 machines on the network so
11:35 these addresses are being used at Network layer so the network layer is going to
11:44 form a packet this packet will have Source IP address which is the address of my
11:51 client machine and this will have destination IP address the address of the server
machine and it will have the
11:59 data from the upper layer transport layer and application layer data so this
particular layer is called
12:09 host to host communication layer so this is going to address a host
12:18 as a client and host as a server now once this packet is
12:25 prepared as I told previously also the communication has to happen vertically down
so from the network layer the data
12:34 is going to get transported to the data link layer now what happens at data link
12:42 layer now at data link layer pdu is called
12:50 frame so the frames are going to get formed at data link layer and this is hardware
dependent
12:58 layer so that that means if I'm using WiFi then the frame format is different if I'm
going to use wired Network the
13:06 frame format is different but whether it is a WiFi network or a wired network one
thing is very common that we'll have a
13:16 source Mac address and destination Mac address so these Mac addresses are
13:23 called physical addresses so IP addresses logical
13:32 address which is at Network layer and Mac address is physical
13:43 address which is at data link layer now if you want to uh just correlate this
13:49 with some analogy so it's very similar that each of you have a mobile phone so your
mobile number is is
13:59 basically a logical number like if your mobile number is a
14:07 10 digit number so this number will represent a logical number like if you change
your uh cell phone and if you
14:16 want to take a new number the same number can be utilized by somebody else so
this is a logical number and then you
14:23 have a physical number which is assigned to your mobile set this is called II
14:32 number so this is your physical address on the same lines on a network we have a
physical
14:39 address and we have a logical address so logical addresses are at Network layer
14:47 and physical addresses are at data link layer so that means when I want to go to
14:54 a data link layer I have to form a
15:02 frame now frame is being formed with the help of source Mac address and
destination Mac

15:11 address now source Mac address or a destination Mac address these Mac addresses are of 48

15:19 bit so this is a 48 bit number which is a hexad decimal notation

15:27 it is defined in a hexad decimal format like 00 colon a0 colon C0 colon

15:38 AB colon CD colon 01 so this is a typical Mac address now this Mac address

15:47 has two parts one is called the manufacturer another is the serial

15:54 number so first three bytes of a MAC address is going to have OEM

16:05 specification so OEM is original equipment manufacturer so the first three bytes represents OEM so this card

16:14 can be from Cisco it may be from Intel it may be from dink so this is the number which will represent that company

16:22 and after that three bytes they will represent the serial number of the net

16:28 network card so that means we have a MAC address which basically gets mapped to the IP

16:36 address which is again going to get ma to the address which is represented in the

16:44 address bar of a browser so this is complete hierarchy which is being followed from an

16:51 application layer to data link layer whenever we want to Traverse or whenever we want to surf internet

16:59 that means from an application layer we specify a string why we specify string because we

17:08 as a human being are very good at remembering names we are not good at remembering numbers but as a machine

17:15 computer system only works on numbers so we need to convert this naming convention into the number convention

17:23 and vice versa the number convention gets translated into naming convention so that that human and machine can work

17:31 together so we specify a string like www do

17:40 google.com now the computer doesn't understand what is www.google.com so it asks me to get some

17:48 logical number for this so This logical number is given at Network layer so

17:55 there will be one IP address which is being given to www.google.com now this IP address will

18:03 finally get converted into a MAC address which is the address being specified on that particular host

18:13 so that particular host will have a 48-bit number and that number will represent the MAC address now as an

18:21 experiment if you want to see what is my IP address and what is my Mac address it's very simple you need to go to the command prompt

18:29 and at the command prompt you enter a command called ip config
18:36 /all now this is going to yield IP address and Mac addresses for your interface like
here in this case if
18:46 you can see this we have an IP address assigned to this machine 1921
18:54 1681 2351 and this is the subnet mask so this is the IP address and subnet
19:01 19 minutes, 1 second mask combination on this particular machine as well as if you
can see
19:06 here this is your physical address 00
19:13 5056 01 so this Mac address is the physical address of this particular machine so
these addresses are unique so
19:23 that means this unique address gets translated into ipv4 address which again gets
translated into
19:31 some name like www.google.com this is how we can see the IP addresses now let's
say I want to
19:39 know what is IP address of www.google.com so you simply type ping
19:46 this is a ping command there is a command whenever you are got connected to the
internet you give command P
19:53 followed by www.google.com now you see it pops up an address now
20:02 you would be very familiar about this address this is 1731 1941 126.80 so this is
the IP
20:12 address which is assigned to www.google.com at this point of time so this IP
address by virtue of this IP
20:22 address I'll be able to make a frame so let's copy this address
20:29 and make a frame copy now let's create a network
20:37 packet first then we are going to form a frame so in the network packet we'll have a
source IP address which is the IP
20:46 address of my machine and then we have a destination IP which is the IP which we
have got
20:55 from Google so this is the IP address which is get going to get translated here in
the destination field so in the
21:04 destination field we'll have this IP address now then this packet goes to the data
link layer now what is the job of a
21:13 data link layer data link layer is going to translate the packet which has come from
a network
21:21 layer and makes a frame out of it now one common mistake which is made
21:29 by students is that the frame Mac address source Mac address is the MAC
21:36 address of the client and destination Mac address is the address of the server
which is not actually the case this
21:45 destination Mac address is the MAC address of your gateway now let's learn this

concept

21:54 that why this destination Mac address is not of the server but this Mac address is of the

22:01 22 minutes, 1 second Gateway now as previously we had done let's take the same example 192.168.1.1 let's say this machine wants

22:10 to talk to 192.168.1.3 machine now the first thing which is

22:18 being done by your computer is it's going to calculate the network address of both of these

22:26 addresses if the network address is same in that case we are going to request the

22:34 physical address from the Machine by way of broadcast let's see how this

22:41 communication will happen

22:44 [Music]

22:48 192.168.1.1 wants to talk to 192.168.1.3 so that mean this machine

22:58 knows about the IP address of another machine so they are at which layer they are at Network

23:06 layer because Source IP is also known and destination IP is also

23:12 known now in order to make a frame I would require a source Mac

23:20 address and I would also require a destination Mac address so the source Mac address will

23:27 be the MAC address add of my machine so I can get it from IP config /all command

23:35 so that Mac address will go here now what about this Mac address now if the machine is on a same network it is going

23:43 to request the MAC address of another Machine by way of

23:52 broadcasting now this broadcasting is called as ARP broadcasting AR RP stands for

24:01 24 minutes, 1 second address resolution protocol so if machine knows the IP

24:10 address of another machine it can only be at Network layer now because it has to go down to the data link layer in

24:19 order to access the frame so it needs to acquire Mac address and Mac address can be require

24:27 acquired with with the help of ARP which is called address resolution protocol so my machine is going to send a ARP

24:39 broadcast and another machine 1.3 in our case is going to reply back with ARP

24:47 reply which is unicost that means my machine is going

24:55 to say that what is Mac address of 1.3 then 1.3 is going to say back that

25:03 my Mac address is 00 0 0 0 C the whole 48 bit

25:10 number now with the help of this 48-bit number my machine will be able to form a frame and with this Frame I'll be able

25:20 to go to the data link layer so that means we'll specify a string that that

25:28 string is converted into some logical number which is called IP and this IP is going to get converted into a physical

25:37 address now this is the case when we are on a same network now if we are on a different network this Mac address will be the MAC

25:46 address of router so this we are going to explain in the next lecture that what is a

25:52 router for this lecture let's first of all understand what we have learned so far for so that we can recap

26:01 26 minutes, 1 second this so in this lecture we understood that classful addressing scheme wastes

26:08 IPS so we can adapt to cidr notation now this cidr notation is going to help us to lower

26:17 down the wastage of IP addresses so as the number of machines are required those many bits can be chosen from the

26:26 IP address and then we need to calculate the network address and broadcast address so we learned about how to

26:34 calculate the network address broadcast address first host address last host address

26:41 also we found out that after knowing this addresses we need to get a MAC address so that we'll be able

26:51 to prepare the frame so Mac address would be delivered with the help of ARP protocol which is called address

26:59 resolution protocol and this Mac address is of 48 bits now out of these 48

27:07 bits first three bytes that means 24 bits will represent the OEM which has manufactured this

27:16 particular card and then next 24 bits they will represent the sequence number

27:25 sequence number of the card getting produced this is how the machines are going to talk from an application layer where we

27:34 are going to specify just a string and then the network layer in between there is a transport layer which

27:41 is responsible for specifying the process addresses so process addresses were called Port

27:48 addresses and then we have a network layer which is going to address the IP and finally we have a data link layer

27:56 which is going to tell us about Mac addresses so each of these layer transport layer is called process to

28:04 process communication Network layer is called host to host communication and data link layer is

28:12 called node to node communication so with these Concepts we are ready to

28:21 go into the topic which is called security and we learn about how these

- 28:27 addresses are helpful in order to get to a particular host and there are certain terms which will come up like we can
- 28:36 spoof these IPS and how this spoofing is done that is a part of the next lecture
- 28:42 thank you thank you very much